ADITYA MALVIYA

Senior Data Engineer | Azure Databricks • PySpark • Delta Lake • Azure Data Factory • Unity Catalog

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PROFESSIONAL SUMMARY

Senior Data Engineer with ~5 years at Tata Consultancy Services (TCS) building and optimizing enterprise Lakehouse platforms on Azure and GCP for Rolls-Royce Digital (aerospace) and Avery Dennison (US Fortune 500). Specialized in Apache Spark performance tuning, Spark Structured Streaming, Delta Lake, Azure Data Factory, and Unity Catalog governance, plus containerized FastAPI data services. Reduced a critical Spark job's runtime by 73% (52 min to 14 min) and Azure platform costs by ~40% while leading a 4-engineer delivery team.

TECHNICAL SKILLS

Languages & Formats: Python, SQL, T-SQL, YAML, Parquet, Delta, JSON, CSV

Spark & Lakehouse: Apache Spark (PySpark, Spark SQL, Structured Streaming), Azure Databricks, Delta Lake (MERGE, OPTIMIZE, Z-ORDER, Auto Loader), Unity Catalog (RBAC, Lineage), Databricks Jobs & Pipelines, Apache Kafka

Architecture & Performance: Medallion Architecture (Bronze, Silver, Gold), Dimensional Modeling (Star Schema, SCD Type 2), Incremental Loads, Data Quality Checks, Adaptive Query Execution (AQE), Key Salting, Skew Handling, Partitioning, Cluster Right-Sizing

Azure & GCP: Azure Data Factory (ADF), Azure Data Lake Storage (ADLS) Gen2, Azure Event Hubs, Azure Blob Storage, Azure Key Vault (Secret Scopes), Service Principals, Azure API Management, Google Cloud Platform (GCP): Databricks on GCP, Vertex AI, Cloud Functions

DevOps & Backend: Azure DevOps CI/CD (YAML Pipelines, Boards), Git, GitHub Actions, Docker, FastAPI, Django, REST APIs, Redis, Pytest, SonarQube, Agile/Scrum

Databases, BI & AI: Microsoft SQL Server, DuckDB (incl. DuckDB-WASM), Power BI, dbt, Apache Airflow 2, Vertex AI Vector Search, ChromaDB, Hybrid RAG (BM25 + Embeddings), WebGPU LLM Inference (WebLLM)

PROFESSIONAL EXPERIENCE

Tata Consultancy Services (TCS) | Mumbai, India — Dec 2021 – Present

Senior Data Engineer (Mar 2026 – Present) | **Data Engineer** (Dec 2021 – Mar 2026)

Client: Rolls-Royce Digital (Aerospace) — *Feb 2022 – Present*

Engine History Analysis – Azure Lakehouse Platform

- Architected and owned ETL pipelines in Databricks Jobs & Pipelines (PySpark, Spark SQL) that process files through a Bronze/Silver/Gold Medallion Lakehouse on Delta Lake and ADLS Gen2, plus 7 production Azure Data Factory (ADF) pipelines for ingestion and ETL/ELT flows, modeling 50–60 Star Schema tables for Power BI.
- Engineered streaming ingestion of 30–40 GB/day with Azure Event Hubs, Apache Kafka, and Spark Structured Streaming, plus 5–7 GB/day of incremental batch loads using Auto Loader and SCD Type 2 Delta MERGE upserts, orchestrated through Databricks Jobs with daily Gold-layer refreshes.
- Reduced a critical PySpark job's runtime by 73% (52 min to 14 min) and eliminated recurring executor out-of-memory (OOM) failures by first applying selective key salting to a skewed join, then enabling and configuring Adaptive Query Execution (AQE) skew-join handling.
- Lowered Azure platform costs by ~40% through right-sized Databricks job clusters, shorter job runtimes, Delta Lake OPTIMIZE and Z-ORDER with table partitioning, and parameterized, reusable ADF pipelines.
- Built Python data quality checks (deduplication, null handling) into the Bronze and Silver layers, governed tables through Unity Catalog (RBAC, data lineage), and secured credentials with Azure Key Vault-backed secret scopes and service principals.
- Delivered ~4 user stories and 2 bug fixes per sprint on feature branches merged into develop and main; prepared Azure DevOps YAML files and ran CI/CD deployments of each branch to Dev for integration testing ahead of DevOps-owned promotion to Stage (limited-user UAT) and Production.
- Led a 4-member offshore team as Scrum Master across 27 two-week sprints and 3 production releases: ran daily stand-ups with UK client stakeholders and sprint planning on Azure DevOps Boards, mentored 4 junior engineers, and reviewed pull requests.

Assumptions Management – Data Services & APIs

- Developed containerized (Docker) FastAPI data services with RBAC and Azure API Management over Microsoft SQL Server and Azure Blob Storage; cut the web app's end-to-end processing time (including pipeline runs) from 5–7 min to 3–4 min via Redis caching and T-SQL query optimization.

- Maintained 95–100% Pytest coverage on every branch and cleared all SonarQube quality-gate findings (reliability, security, maintainability, coverage, cognitive complexity) before merges to develop and main.

Client: Avery Dennison (Smartrac RFID, US Fortune 500) – R&D Data & AI Platform *Dec 2021 – Jan 2022*

- Engineered Databricks pipelines on GCP to ingest, cleanse, and prepare a 10–15 GB batch RFID R&D dataset for ML model training, and built a production RAG document Q&A tool on Vertex AI, Cloud Functions, Vertex AI Vector Search, and ChromaDB that lets researchers upload and query up to 4 documents at once.

BYJU'S (Gradeup) | Remote Jul 2021 – Nov 2021

Python Developer Intern

- Developed Python and Django backend features for Gradeup's exam-preparation web platform.

RESEARCH

AQE vs. Salting: A Cost Model and Decision Procedure for Skewed Joins in Apache Spark Aug 2026

Independent research paper (13 pages) | Paper (PDF) | GitHub

- Built SKEWBENCH, a reproducible Spark 3.5.3 benchmark comparing six join strategies through Spark event logs, with a salt-cardinality cost model: AQE's skew-join rule stayed inactive until advisoryPartitionSizeInBytes fell below the hot-partition size, while selective salting cut the join's critical path 1.9× (single node) and replicated 11 shuffle rows per salt value vs. 20,000 for uniform salting.

PROJECTS

PipeScope – Spark Performance & Column-Level Lineage Analyzer | Live | GitHub Aug 2025 – Present

- Built a browser-based Spark diagnostics tool that parses Spark JSON event logs in a Web Worker to flag 15 issue types (skew, disk spill, GC pressure, small files, disabled AQE) with copy-ready fixes, and maps SQL/dbt column-level lineage via sqlglot on Pyodide; deployed through GitHub Actions.

FlowForge – Visual ETL Builder with PySpark & dbt Code Export | Live | GitHub Oct 2026

- Built a visual ETL builder with 12 step types (joins, window functions, deduplication, data quality checks) that re-runs every change live on DuckDB-WASM and exports the same pipeline graph as Spark SQL, PySpark, DuckDB SQL, and a dbt project with schema tests; deployed through GitHub Actions.

Orchestra – Pipeline Orchestrator & Run Doctor | Live | GitHub Oct 2026

- Designed a deterministic Airflow-style scheduler on a simulated clock with cron catch-up, exponential-backoff retries, pools, four trigger rules, SLAs, and idempotent backfills; its run doctor separates root cause from blast radius, and DAGs export to Apache Airflow 2.

Corpus – Browser-Native Hybrid RAG & Text-to-SQL Workspace | Live | GitHub Oct 2026

- Built a zero-server document and data Q&A app running Gemma 4 E2B on WebGPU (WebLLM) with hybrid BM25 + vector retrieval, MMR re-ranking, intent routing, and cited answers; converts questions on CSV/Parquet files into read-only DuckDB-WASM SQL and scores 100% Hit@1 (hybrid) on an 8-question golden set.

Additional builds, including DataLens (data quality profiling on DuckDB-WASM) and behaviorlab (Kafka event pipeline with Bronze/Silver/Gold layers), are on the portfolio.

CERTIFICATIONS

Microsoft Certified: Fabric Data Engineer Associate (DP-700)
Microsoft Certified: Azure Data Fundamentals (DP-900)

EDUCATION

Bachelor of Technology (B.Tech), Mechanical Engineering | Bhilai Institute of Technology, Durg, India Aug 2021